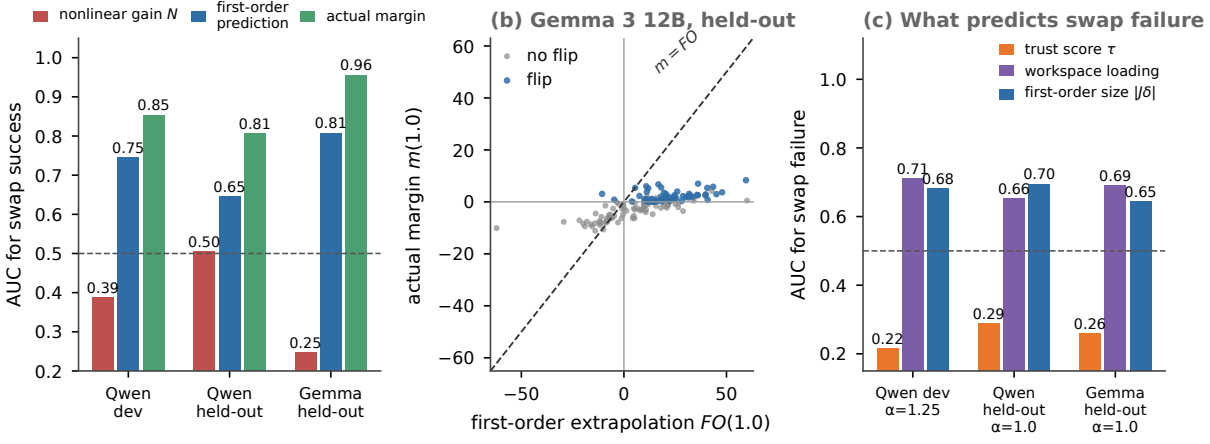
(a) What predicts swap success
nonlinear gain N
first-order prediction
actual margin
AUC for swap success
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.39
0.75
0.85
0.50
0.65
0.81
0.25
0.81
0.96
Qwen dev
Qwen held-out
Gemma held-out
(b) Gemma 3 12B, held-out
no flip
flip
m = FO
actual margin m(1.0)
60
40
20
0
−20
−40
−60
−50
0
50
first-order extrapolation FO(1.0)
(c) What predicts swap failure
trust score τ
workspace loading
first-order size |Iδ|
AUC for swap failure
1.0
0.8
0.6
0.4
0.2
0.22
0.71
0.68
0.29
0.66
0.70
0.26
0.69
0.65
Qwen dev α=1.25
Qwen held-out α=1.0
Gemma held-out α=1.0